

# Multiple Hypothesis Testing (in DEG, GWAS)

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# An example: Genome-wide Association Studies

## GWAS

- Input:

- A genotype matrix  $X$  ( $n \times p$ ), where  $X_{ij} \in \{0, 1, 2\}$
- A phenotype vector  $\mathbf{y}$  ( $n \times 1$ )

- Optimization:

$$\mathbf{y} = \mathbf{x}_j \beta_j + \epsilon, \text{ for each } j$$

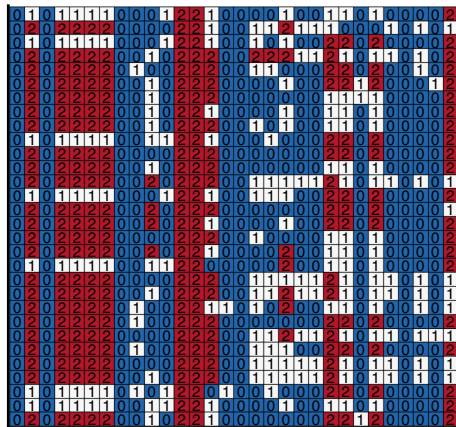
- Output:

- Estimate an effect size (coefficient)  $\beta_j$  ( $j \in 1, \dots, p$ )

- Testing  $H_0 : \beta_j = 0$  vs.  $H_1 : \beta_j \neq 0$ .

# GWAS to map locations associated with case vs. control tests

A genotype matrix  $X$  ( $X_{ij} \in \{0, 1, 2\}$ ):



A vector of phenotypes  $Y$  ( $Y_i = 1$  if case vs.  $Y_i = 0$  if control)

```
table(Y)
```

```
## Y
```

```
##      0      1
```

```
## 1272 1218
```

Goal: Find locations (columns of  $X$ ) where a genotype  $x_j$  is associated with a phenotype  $y$

# GWAS by variant-by-variant association tests

A variant-by-variant association t-test for a variant  $j$ :

$$H_0 : \mathbb{E}[X_{ij}|Y_i = 1] = \mathbb{E}[X_{ij}|Y_i = 0] \quad \text{vs.}$$

$$\mathbb{E}[X_{ij}|Y_i = 1] \neq \mathbb{E}[X_{ij}|Y_i = 0]$$

- The average genotype is the same between the case and control under the null.

```
## library(matrixTests)
.gwas.file <- "../data/genotype/gwas_results.rds"
if.needed(.gwas.file, {
  .gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
  saveRDS(.gwas, .gwas.file)
})
```

# Today's lecture

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- 1 Hypothesis testing in terms of two errors
- 2 Controlling Family-Wise Error Rate
- 3 False Discovery Rate
- 4 FWER, FDR, then what?
- 5 Beyond global FDR: gene-by-gene localized control

# Recap: Steps of hypothesis testing

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## ① Choose a test statistic and formulate hypotheses

- Null hypothesis  $H_0$ : no effect / no difference

## ② Define the rejection region

- Set significance level  $\alpha$  (e.g., 0.05)
- Determine critical values or thresholds

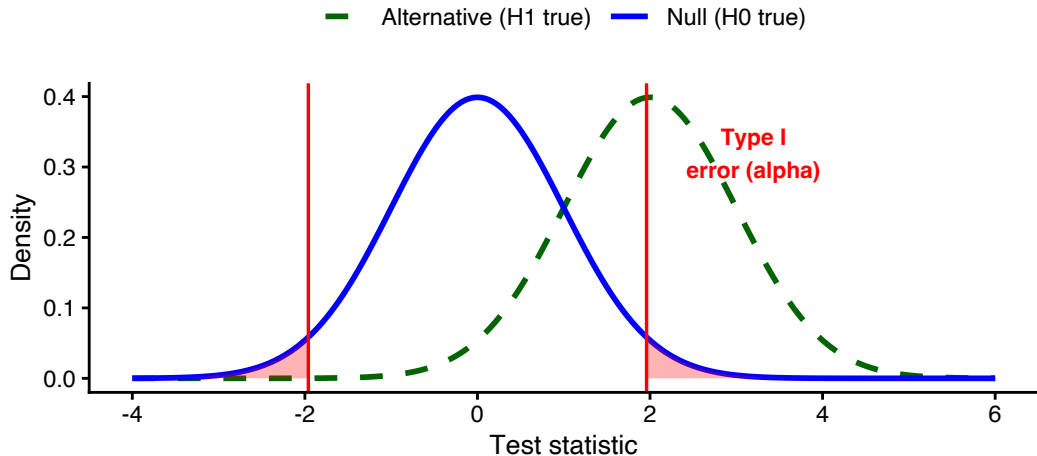
## ③ Compute the test statistic from observed data

## ④ Make a decision

- If test statistic falls in rejection region (or p-value  $< \alpha$ ): reject  $H_0$

# What does type I error mean for a single test?

Type I error: rejecting  $H_0$  when  $H_0$  is true



# Type I and Type II errors (lect04)

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|                                | $H_0$ True                | $H_0$ False               |
|--------------------------------|---------------------------|---------------------------|
| <b>Reject <math>H_0</math></b> | Type I Error ( $\alpha$ ) | Correct (Power)           |
| <b>Fail to reject</b>          | Correct                   | Type II Error ( $\beta$ ) |

- **Type I Error** (false positive): Reject  $H_0$  when it is actually true
- **Type II Error** (false negative): Fail to reject  $H_0$  when it is actually false

# Statistical power: $1 - \beta$

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**Power** = Probability of correctly rejecting  $H_0$  when it is false

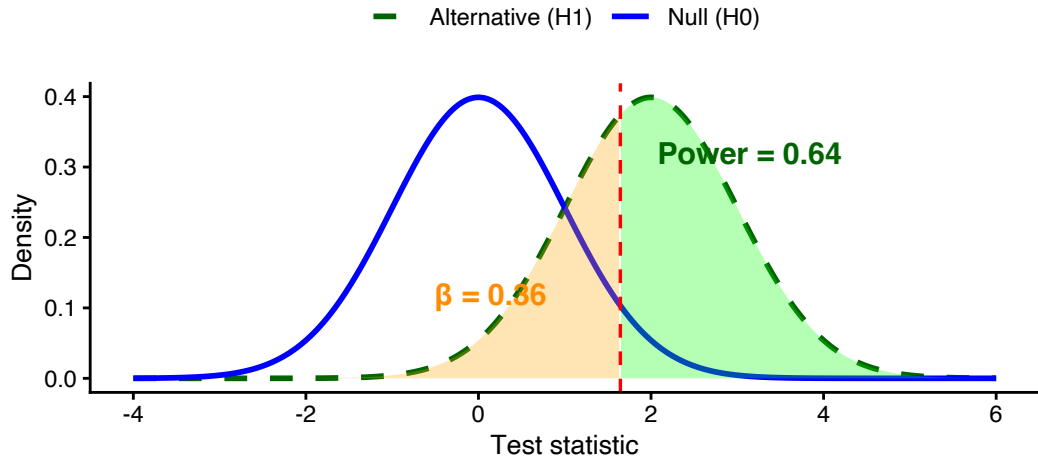
$$\text{Power} = 1 - \beta = P(\text{reject } H_0 \mid H_1 \text{ is true})$$

Power depends on:

- 1 **Significance level**  $\alpha$  (Type I error rate)
- 2 **Effect size**  $\delta$  (how different  $H_1$  is from  $H_0$ )
- 3 **Sample size**  $n$
- 4 **Variability**  $\sigma$

# Visualizing power

Power analysis:  $\alpha = 0.05$ , effect size = 2.0 SD



# Why statistical power changes with $n$

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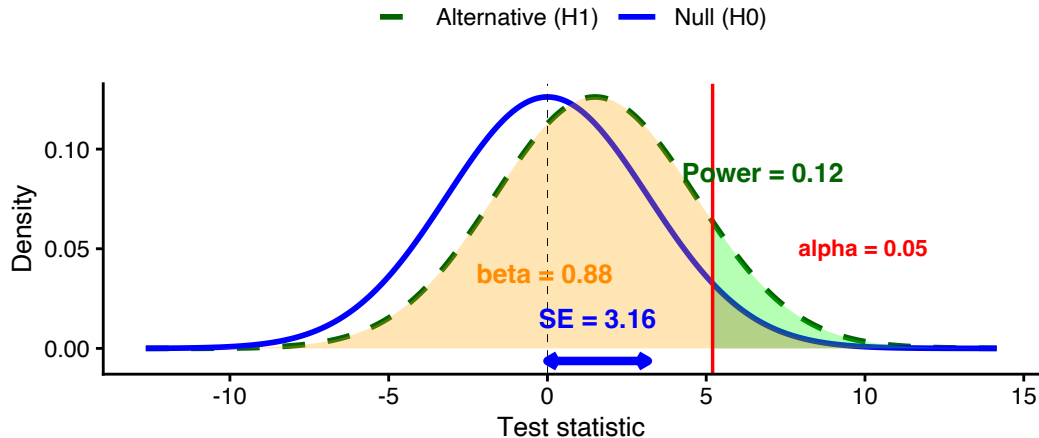
**Key insight:** As  $n \uparrow$ , the standard error  $\downarrow$

$$SE = \frac{\sigma}{\sqrt{n}}$$

- **Both** null and alternative distributions have the same SE
- Larger  $n \rightarrow$  **both distributions become narrower**
- Fixed effect size  $\delta \rightarrow$  distributions **separate more** (relative to SE)
- Less overlap  $\rightarrow$  easier to distinguish  $H_0$  from  $H_1$
- Result: **higher power**

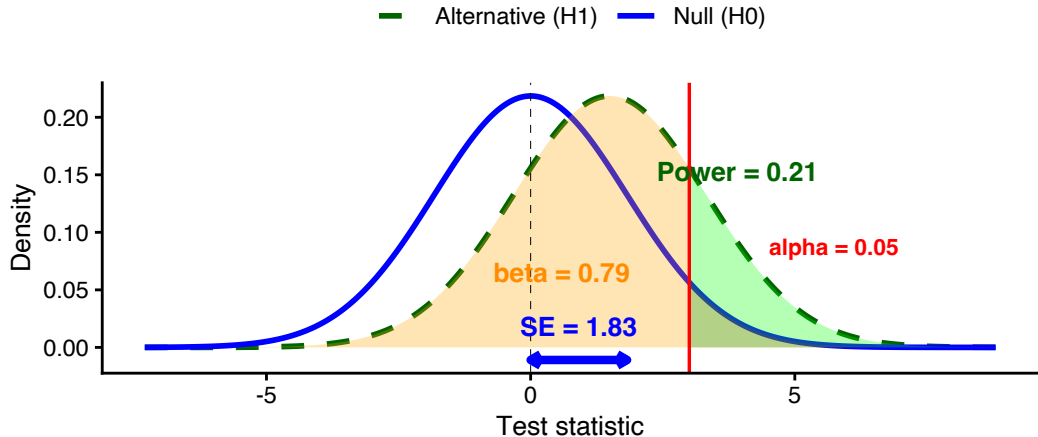
# Power increases with sample size: visualization

$n = 10$



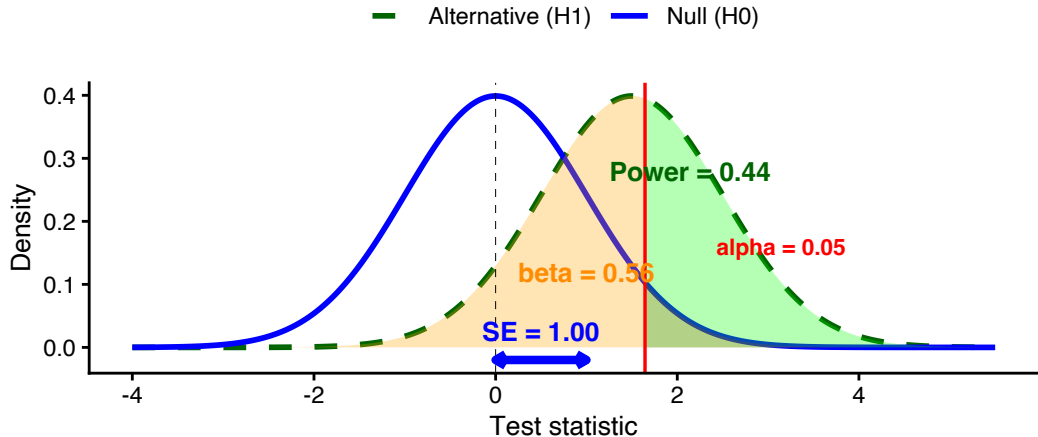
# Power increases with sample size: visualization

$n = 30$



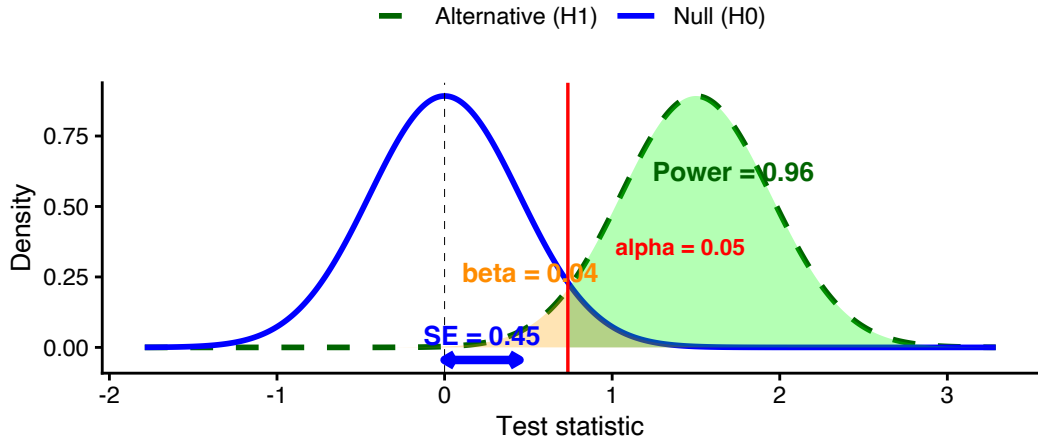
# Power increases with sample size: visualization

$n = 100$



# Power increases with sample size: visualization

$n = 500$



# What do we need to perform power calculation?

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## ① Significance level ( $\alpha$ )

- Type I error rate we're willing to accept

## ② Effect size ( $\delta$ or Cohen's $d$ )

- Can estimate from pilot data or literature

## ③ Sample size ( $n$ )

- Number of observations per group

## ④ Standard deviation ( $\sigma$ )

- Estimate from prior experiments

## ⑤ Power ( $1 - \beta$ )

- Higher power = more confident in detecting true effects

## Typical questions:

- Given  $\alpha$ ,  $\delta$ ,  $\sigma$ , and desired power  $\rightarrow$  find  $n$
- Given  $\alpha$ ,  $n$ ,  $\sigma$ ,  $\delta \rightarrow$  find power

# Power calculation in R (e.g., t-test)

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**Question:** How much power do we have with the sample size  $n$ ?

```
power.t.test(  
  delta = 0.5,      # Effect size (mean diff)  
  sd = 1,           # Standard deviation  
  n = 50,           # Sample size per group  
  sig.level = 0.05, # Rejection region (desired Type I)  
  type = "two.sample"  
)
```

**Answer:** roughly 0.7 power

# Sample size calculation

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**Question:** How many samples do we need to achieve 80% power?

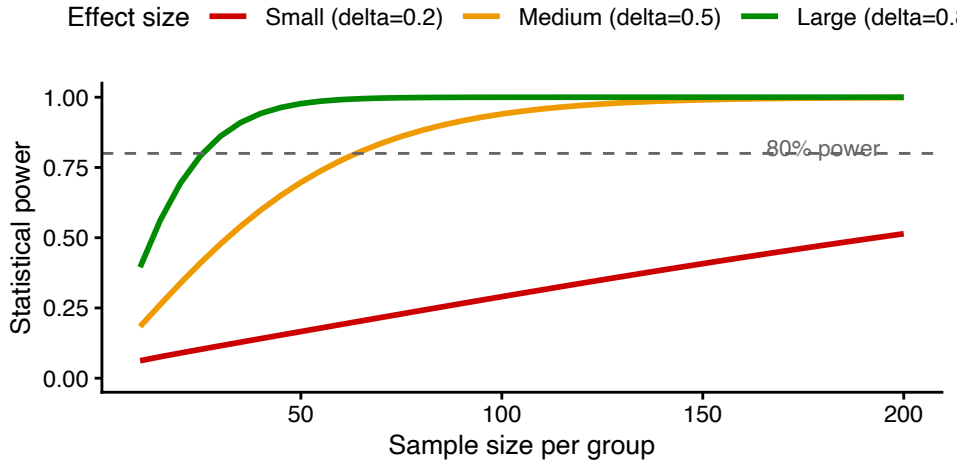
*# Calculate required sample size for 80% power*

```
power.t.test(  
  delta = 0.5,      # Effect size...  
  sd = 1,           # Standard deviation  
  power = 0.8,      # Desired power  
  sig.level = 0.05,  
  type = "two.sample"  
)
```

**Answer:**  $\approx 64$  samples per group for the effect size  $\delta = 0.5$

# Power curves: sample size vs. power

Power curves for two-sample t-test ( $\alpha = 0.05$ )

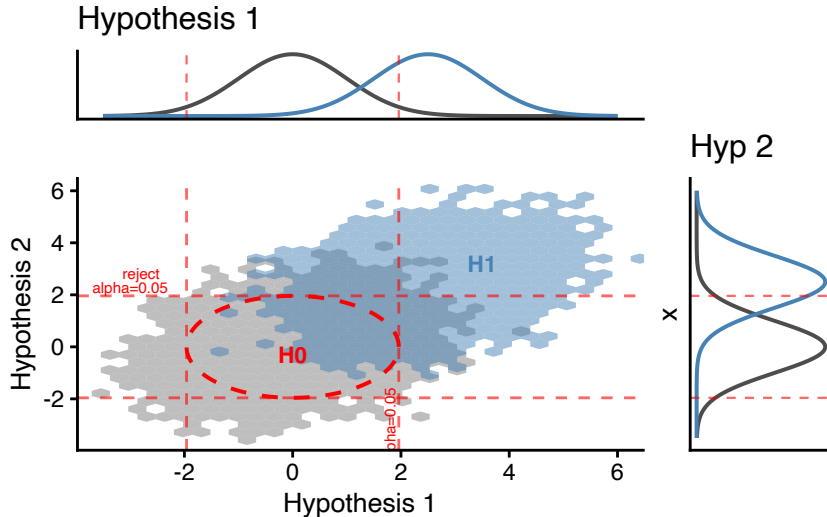


# Today's lecture

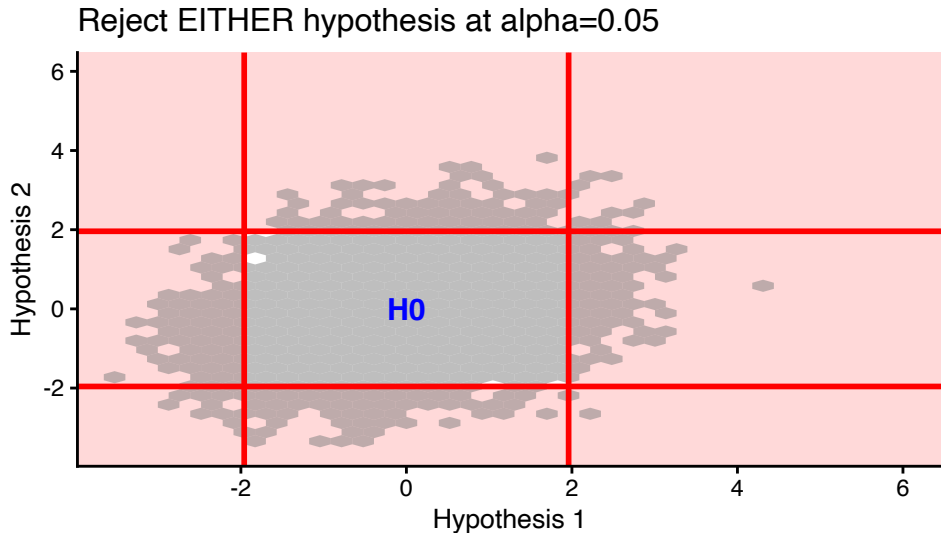
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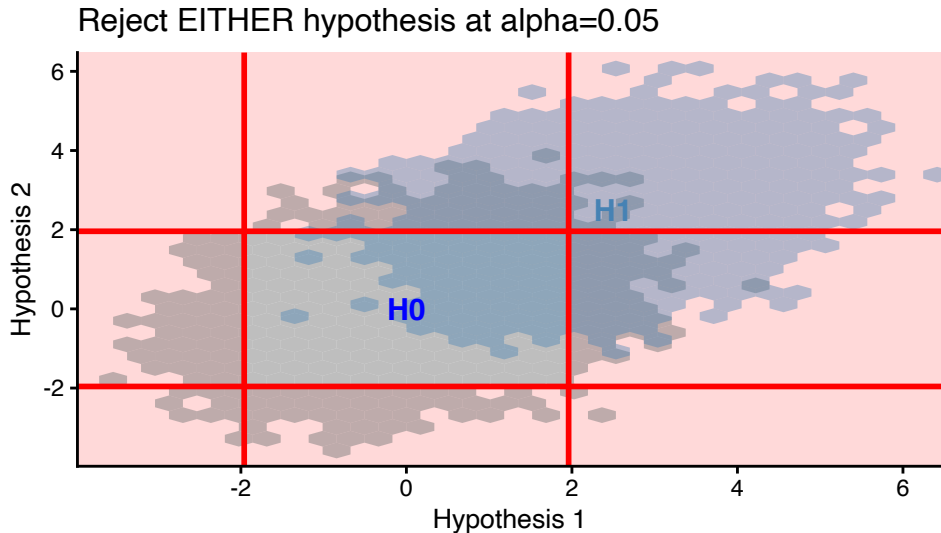
# What if we have more than one hypothesis?



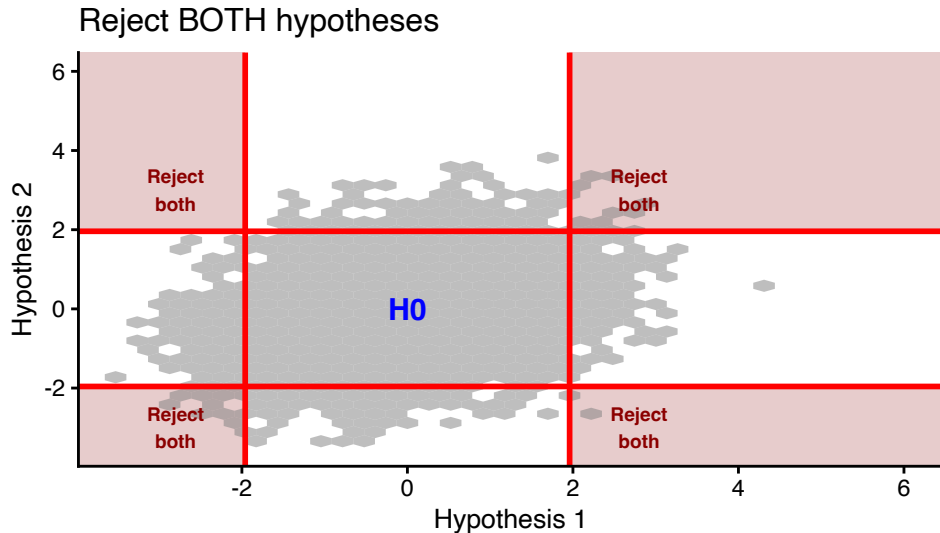
# What if we decided to reject either one at $\alpha < 0.05$



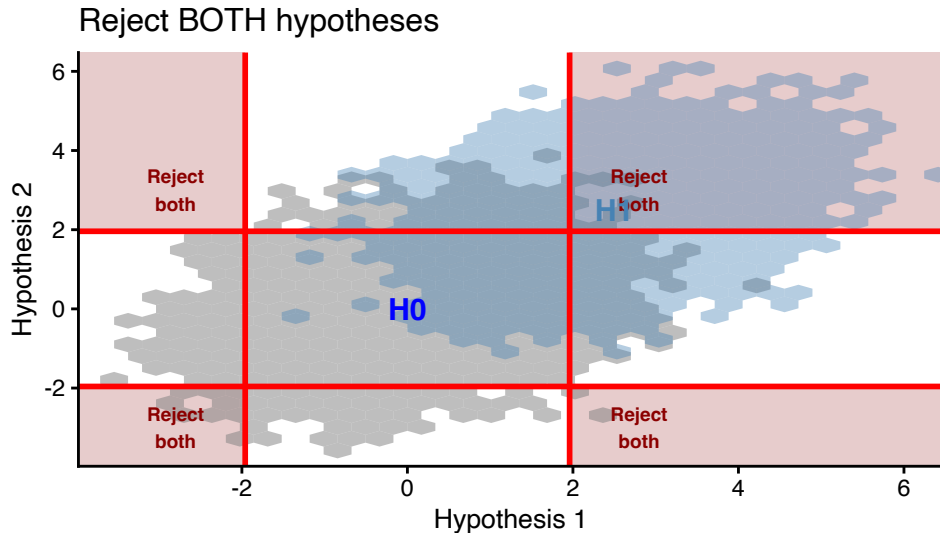
# What if we decided to reject either one at $\alpha < 0.05$



# What if we test each hypothesis $\alpha < 0.05$ ?



# What if we test each hypothesis $\alpha < 0.05$ ?



# Wait, what was the definition of p-value?

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**P-value:** Probability of observing a test statistic as extreme or more extreme than what we observed, **assuming the null hypothesis is true**

$$P(\text{data as extreme or more} \mid H_0 \text{ is true})$$

For a single test at  $\alpha = 0.05$ :

- $P(\text{Type I error}) = P(\text{reject } H_0 \mid H_0 \text{ is true}) < \alpha = 0.05$

# What about two p-values?

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**Union rule** (probability of A or B):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) < P(A) + P(B)$$

**Independence** (probability of A and B when independent):

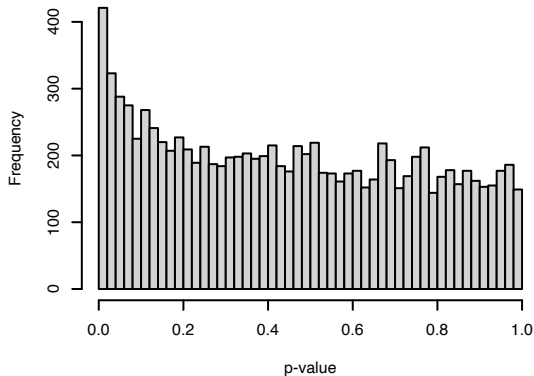
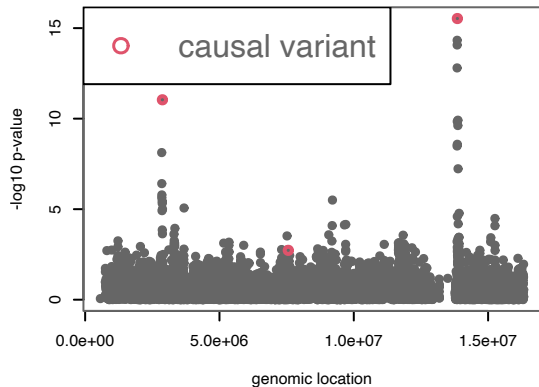
$$P(A \cap B) = P(A) \times P(B)$$

When we want to “control” two hypothesis testing....

$$P(\text{reject at least one true } H_0 \mid H_0) \leq \alpha + \alpha$$

**What will be the  $\alpha$  value that is guaranteed to control both?**

# Manhattan plot to show all the p-values...



We have  $p = 10,000$  p-values. How do we decide which variants are significant?

# The problem: we tested 10,000 hypotheses

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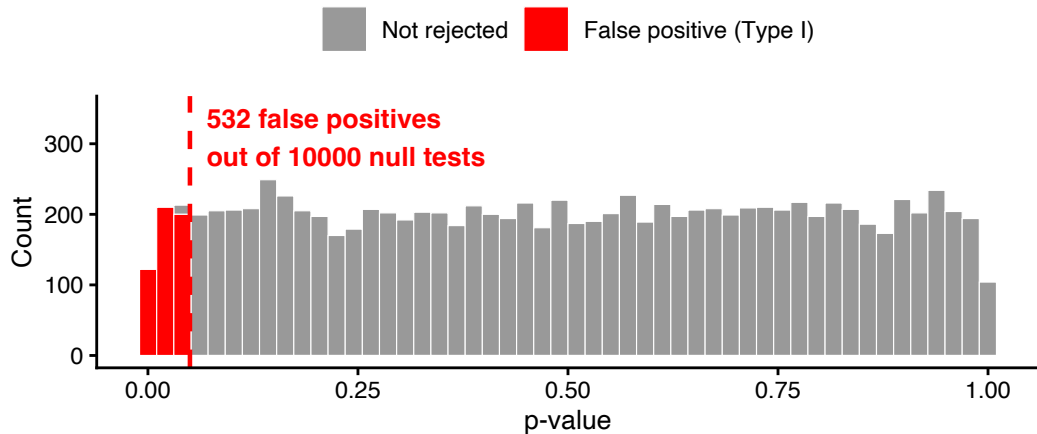
- We performed 10,000 simultaneous tests.
- At  $\alpha = 0.05$ , how many false positives do we expect **by chance alone**?

$$\mathbb{E}[\# \text{ false positives}] = 10,000 \times 0.05 = 500$$

- We need a principled way to control errors when testing many hypotheses at once.

# Why does this matter for 10,000 tests?

All 10,000 tests are null — yet we get ~500 'significant' results



## Conversely.... The dimensionality works unfavourably...

Each test at  $\alpha = 0.05$ : 5% chance of a false rejection if  $H_0$  is true.

| Variants ( $m$ ) | $P(\geq 1 \text{ false rejection})$ |                |
|------------------|-------------------------------------|----------------|
| 1                | $1 - 0.95^1 = 0.05$                 | Just $\alpha$  |
| 2                | $1 - 0.95^2 = 0.098$                | Nearly doubled |
| 3                | $1 - 0.95^3 = 0.143$                | Nearly tripled |
| 10               | $1 - 0.95^{10} = 0.40$              | 40% chance!    |
| 100              | $1 - 0.95^{100} = 0.994$            | Almost certain |
| 10,000           | $1 - 0.95^{10000} \approx 1$        | Guaranteed     |

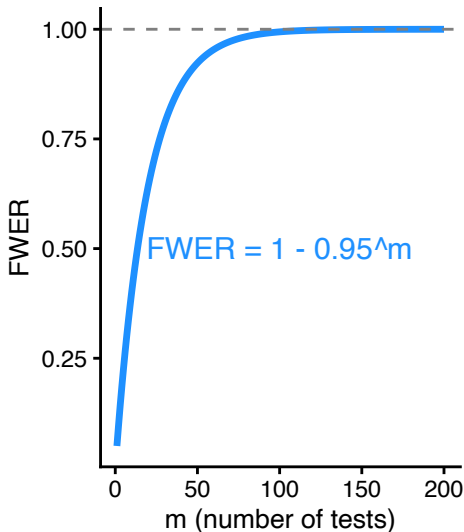
# Family-wise error rate (FWER)

- $P(\text{incorrect} \mid H_0) = \alpha$ ; let  $\alpha = 0.05$
- $P(\text{correct} \mid H_0) = 1 - \alpha = 0.95$
- For  $m$  **independent** tests, all under  $H_0$ :

$$P(\text{all correct}) = (1 - \alpha)^m = 0.95^m$$

- **Family-wise error rate:**

$$\text{FWER} = P(\text{at least one error}) = 1 - 0.95^m$$



# Bonferroni correction: controlling the FWER

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By the **union bound**:

$$P(\text{at least one error}) \leq \sum_{i=1}^m P(\text{error on test } i)$$

Set each test's threshold to  $\alpha_{\text{Bon}} = \frac{\alpha}{m}$ :

$$\sum_{i=1}^m \alpha_{\text{Bon}} = m \cdot \frac{\alpha}{m} = \alpha$$

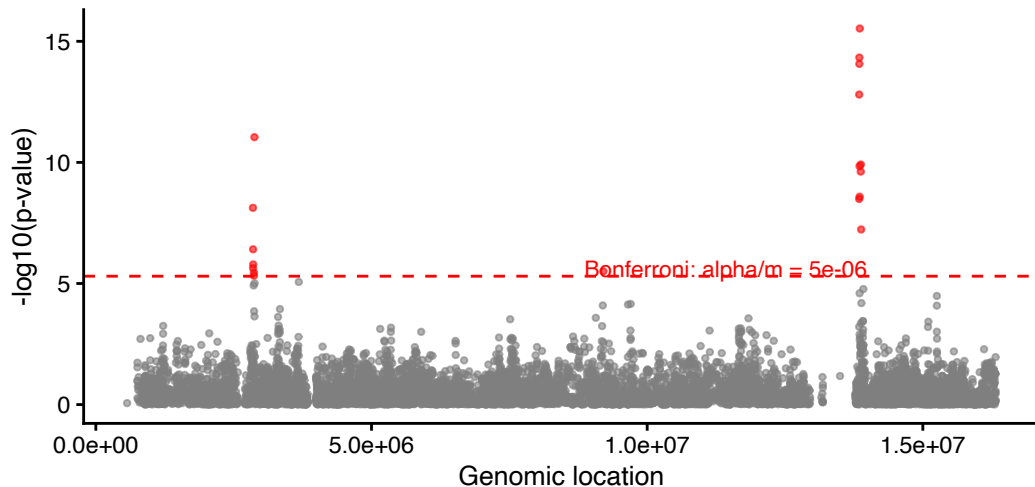
# Bonferroni correction: controlling the FWER

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- Reject  $H_{0,i}$  if  $p_i < \alpha/m$ , or equivalently,
- Adjust  $p_{\text{Bon},i} = \min(m \cdot p_i, 1)$  and threshold at  $\alpha$ .

# Bonferroni on our GWAS example

Bonferroni: 18 rejections out of 10000 tests



# What if we remove correlated variants (LD clumping)?

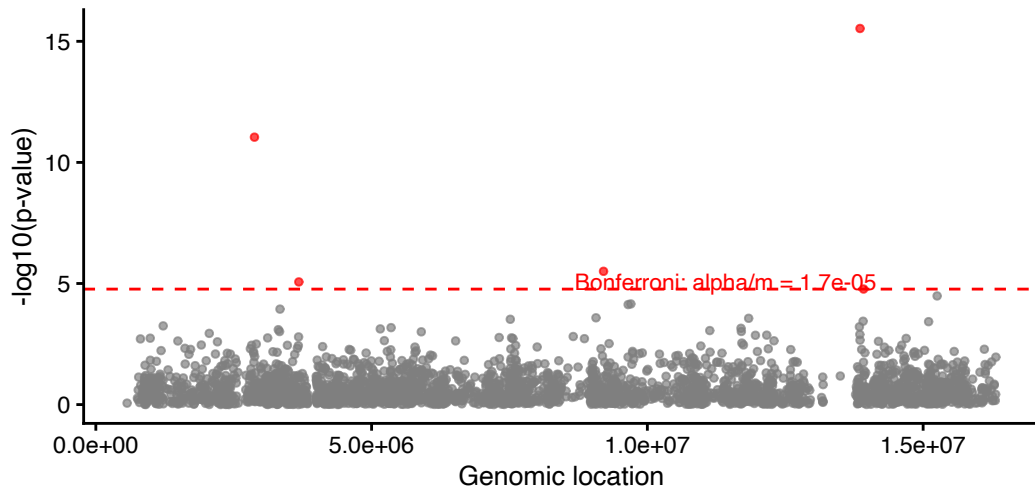
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**After LD clumping, FWER methods work much better!**

- **LD clumping** removes correlated variants, keeping only representative independent SNPs
  - It still takes some threshold parameter...
- Independent tests  $\Rightarrow$  FWER and FDR are better aligned
- But: we lose power by discarding many variants

# FWER control with LD clumping

After LD clumping: 5 rejections out of 2934 tests



# Holm–Bonferroni: a less conservative FWER control (Holm 1979)

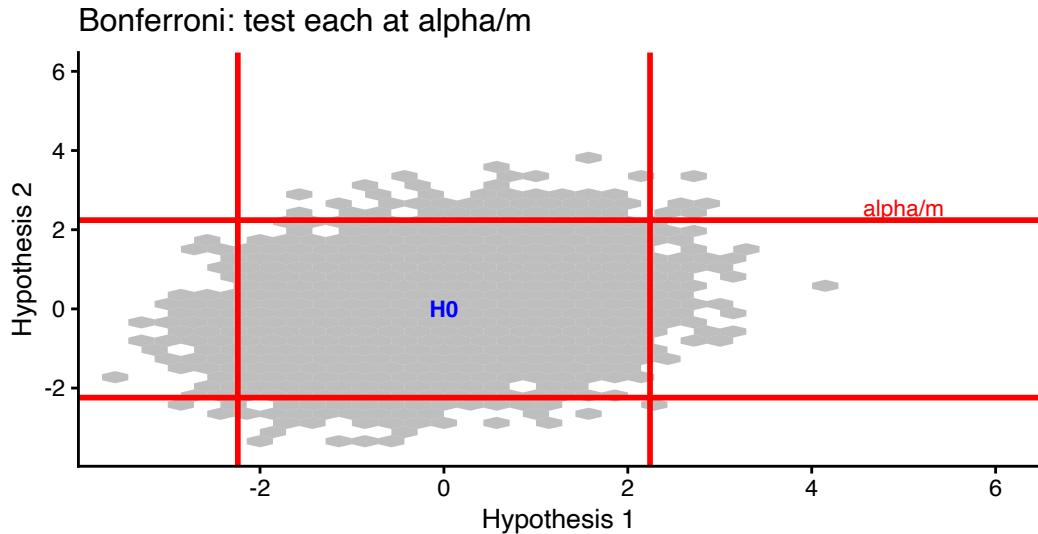
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- **Holm's step-down** procedure uses decreasing multipliers:

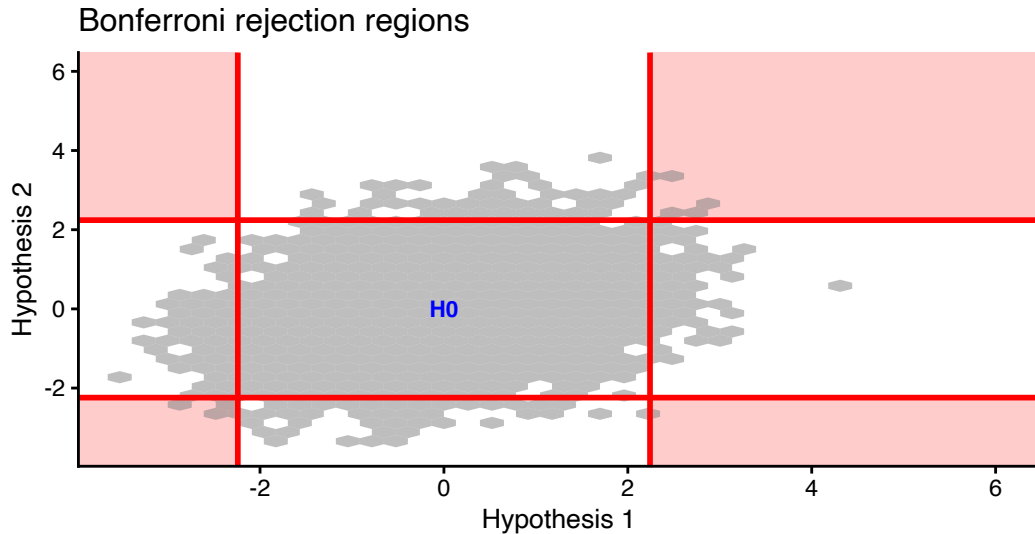
$$p_{\text{Holm}(1)} = m \cdot p_{(1)}, \quad p_{\text{Holm}(2)} = (m - 1) \cdot p_{(2)}, \quad \dots$$

- Still controls  $\text{FWER} \leq \alpha$ , but is **uniformly more powerful** than Bonferroni
- In R: `p.adjust(p, method = "holm")`

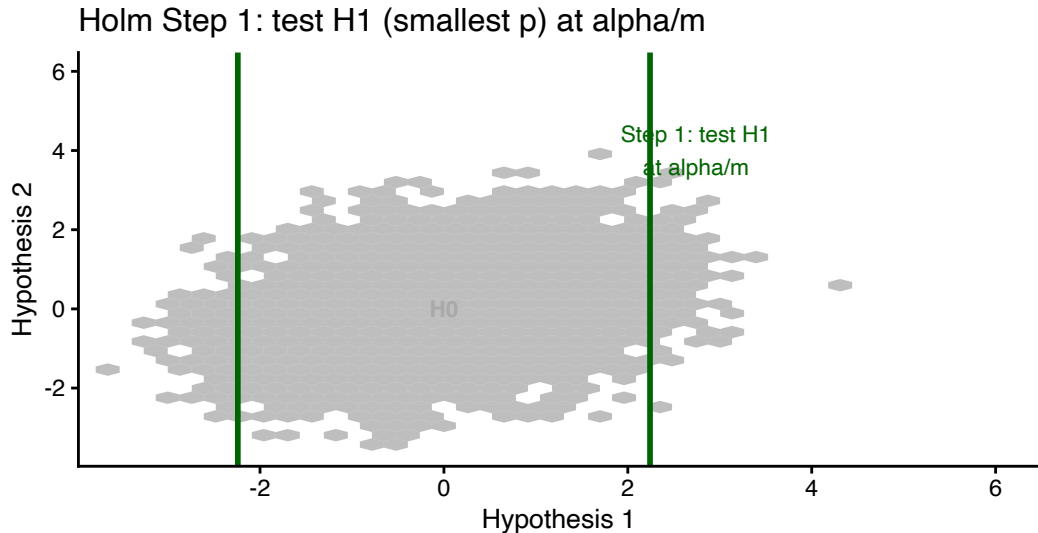
# Bonferroni: uniform strict threshold



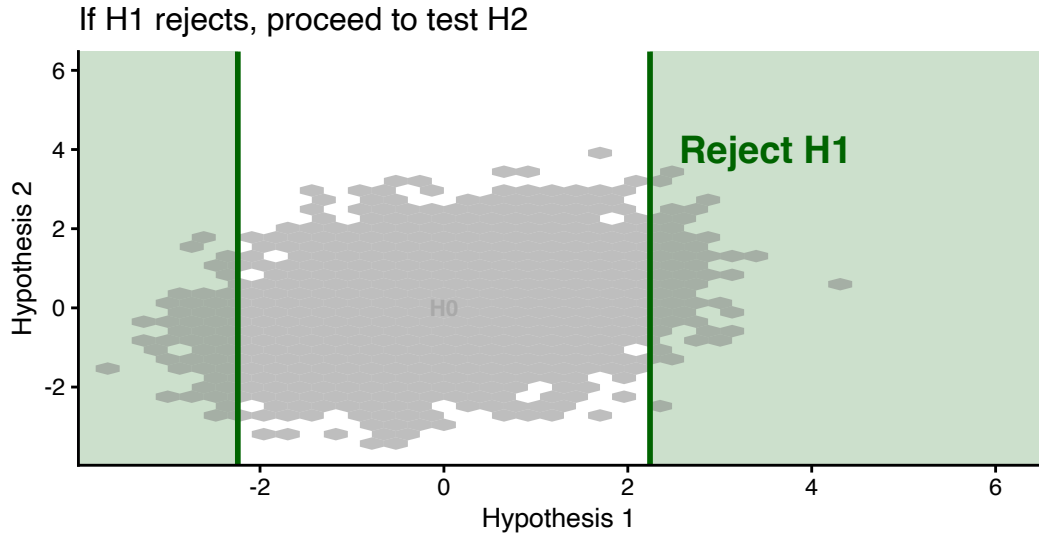
# Bonferroni: uniform strict threshold



# Holm: sequential testing, one axis at a time

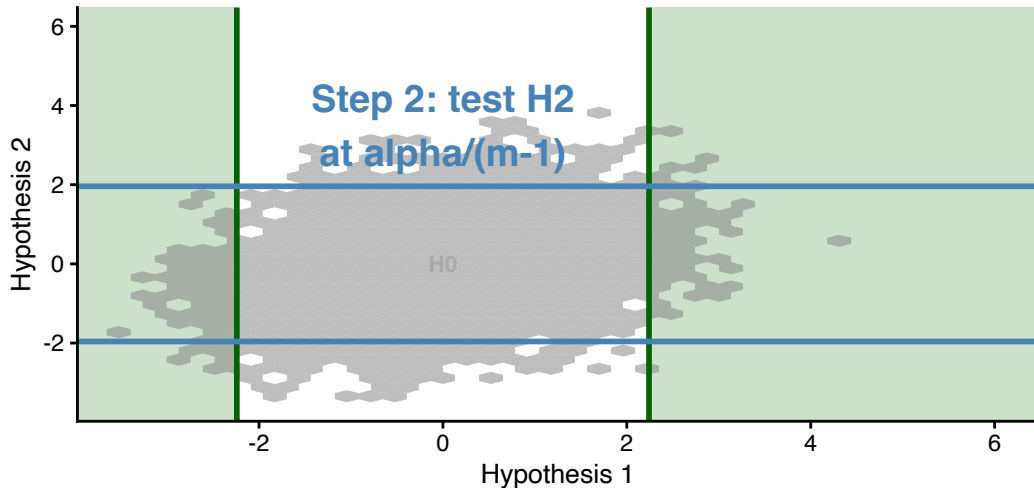


# Holm: sequential testing, one axis at a time



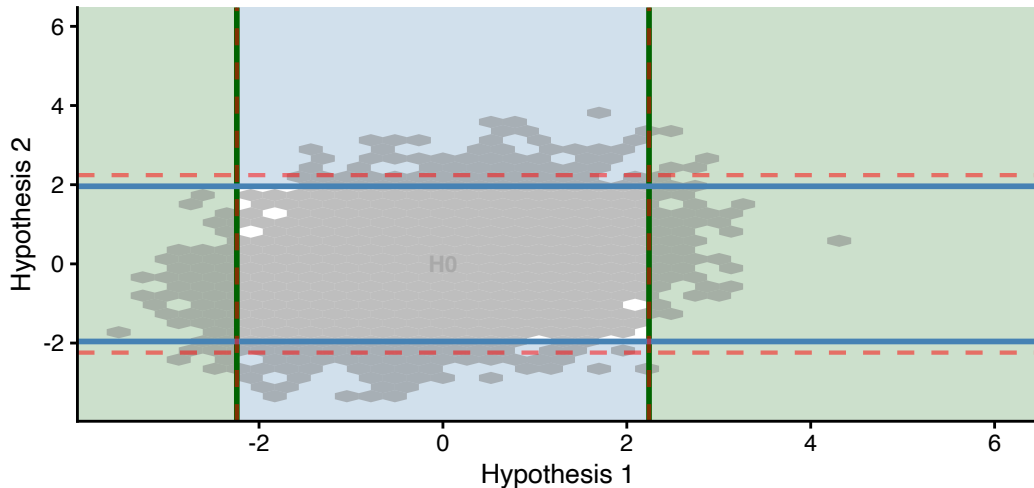
# Holm: sequential testing, one axis at a time

Holm Step 2: test H2 at relaxed threshold  $\alpha/(m-1)$



# Holm: sequential testing, one axis at a time

Holm: sequential rejection regions (axis-by-axis)



# Why is FWER impractical for genomics?

- So much about making **one single error** out of 20,000 tests?
- One easy way to guarantee no Type I errors: **reject nothing!**
  - But...

## The trade-off

Being overly strict about Type I error  $\Rightarrow$  greater Type II error.

- **Key idea:** in high-throughput biology, it is acceptable to make some errors, as long as we have a *comparatively* large number of true positives.

## Pop-up quiz

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Why do we use  $5 \times 10^{-8}$  as a GWAS significance threshold?

# Today's lecture

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# The multiple testing table — not the probability

|             | Not rejected | Rejected | Total |
|-------------|--------------|----------|-------|
| $H_0$ true  | $U$          | $V$      | $m_0$ |
| $H_0$ false | $T$          | $S$      | $m_1$ |
| Total       | $m - R$      | $R$      | $m$   |

- $R$  = total rejections (observed)
- $V$  = false positives (unobserved)
- $S$  = true positives (unobserved)

# FDR vs. FPR vs. FWER

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- **False Discovery Rate (FDR):**

$$\text{FDR} = \mathbb{E} \left[ \frac{V}{R} \right] = \mathbb{E} \left[ \frac{\text{\#false discoveries}}{\text{\#rejections}} \right] \quad (\text{define as 0 when } R = 0)$$

- **False Positive Rate (FPR):**
- **Family-Wise Error Rate (FWER):**

# FDR vs. FPR vs. FWER

---

- **False Discovery Rate (FDR):**
- **False Positive Rate (FPR):**

$$\text{FPR} = \mathbb{E} \left[ \frac{V}{m_0} \right] = \mathbb{E} \left[ \frac{\text{\#false discoveries}}{\text{true null}} \right]$$

- **Family-Wise Error Rate (FWER):**

# FDR vs. FPR vs. FWER

---

- **False Discovery Rate (FDR):**
- **False Positive Rate (FPR):**
- **Family-Wise Error Rate (FWER):**

$$\text{FWER} = P(V \geq 1) = P(\text{at least 1 false discovery})$$

# Benjamini–Hochberg (BH) procedure (Benjamini and Hochberg 1995)

Control FDR at level  $q$  (e.g.  $q = 0.05$ ):

- 1 Order the raw p-values:  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$
- 2 Find the largest rank  $j$  such that  $p_{(j)} \leq \frac{j \cdot q}{m}$
- 3 Reject all hypotheses with rank  $\leq j$

## Adjusted p-values

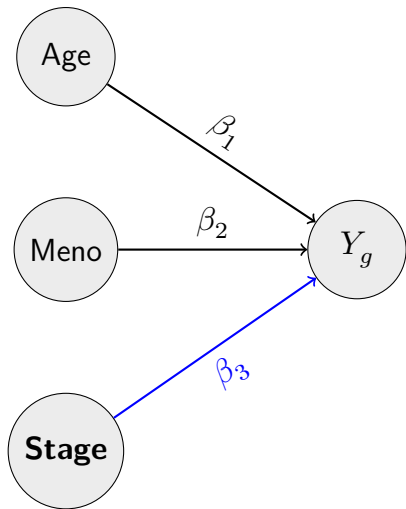
$$p_{(i)}^{\text{BH}} = \min_{j \geq i} \left( \frac{m}{j} p_{(j)} \right), \quad \text{capped at 1}$$

## BH worked example ( $m = 10, q = 0.05$ )

| Rank $j$ | P-value | BH threshold $\frac{jq}{m}$ | Adjusted p-value $p^{\text{BH}}$ | Reject? |
|----------|---------|-----------------------------|----------------------------------|---------|
| 1        | 0.0008  | 0.005                       | 0.008                            | ✓       |
| 2        | 0.009   | 0.010                       | 0.045                            | ✓       |
| 3        | 0.127   | 0.015                       | 0.423                            |         |
| 4        | 0.205   | 0.020                       | 0.513                            |         |
| 5        | 0.396   | 0.025                       | 0.750                            |         |
| 6        | 0.450   | 0.030                       | 0.750                            |         |
| 7        | 0.641   | 0.035                       | 0.916                            |         |
| 8        | 0.781   | 0.040                       | 0.976                            |         |
| 9        | 0.900   | 0.045                       | 0.993                            |         |
| 10       | 0.993   | 0.050                       | 0.993                            |         |

Bonferroni ( $\alpha/m = 0.005$ ) would only reject rank 1.

# DEG: expression $Y \sim$ cancer stage in TCGA BRCA



Linear model for gene  $g$ :

$$Y_{ig} = \beta_{1g} \cdot \text{Age}_i + \beta_{2g} \cdot \text{Meno}_i + \beta_{3g} \cdot \text{Stage}_i + \epsilon_{ig}$$

where  $\epsilon_{ig} \sim N(0, \sigma_g^2)$

Null hypothesis for gene  $g$ :

$$H_{0g} : \beta_{3g} = 0$$

Testing  $m = 20,000$  genes  $\Rightarrow m = 20,000$  hypotheses!

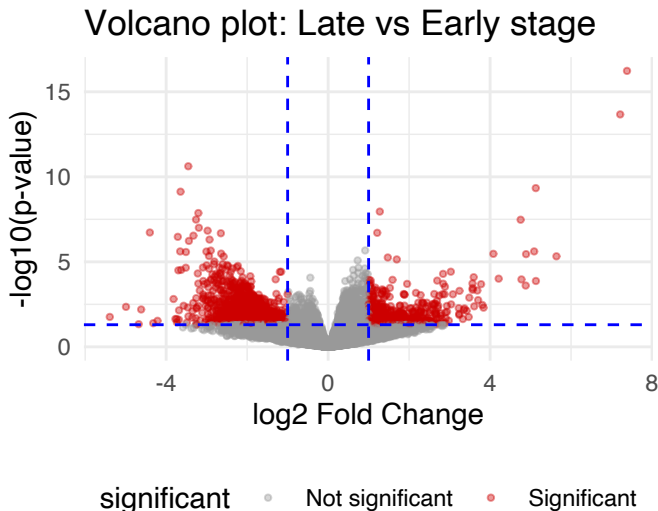
# Run limma-voom analysis

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| ## |                    | logFC     | AveExpr   | t         | P.Value      | adj.P.Val    |
|----|--------------------|-----------|-----------|-----------|--------------|--------------|
| ## | ENSG00000221867.8  | 7.381752  | -8.937562 | 8.505772  | 5.869385e-17 | 3.293606e-12 |
| ## | ENSG00000197172.10 | 7.213459  | -9.570496 | 7.747667  | 2.130298e-14 | 5.977083e-10 |
| ## | ENSG00000233735.5  | -3.453995 | -9.963162 | -6.750199 | 2.392598e-11 | 4.475355e-07 |
| ## | ENSG00000198930.12 | 5.128308  | -7.603639 | 6.289275  | 4.605533e-10 | 6.460987e-06 |
| ## | ENSG00000233473.2  | -3.645117 | -9.021456 | -6.211074 | 7.468179e-10 | 8.381537e-06 |

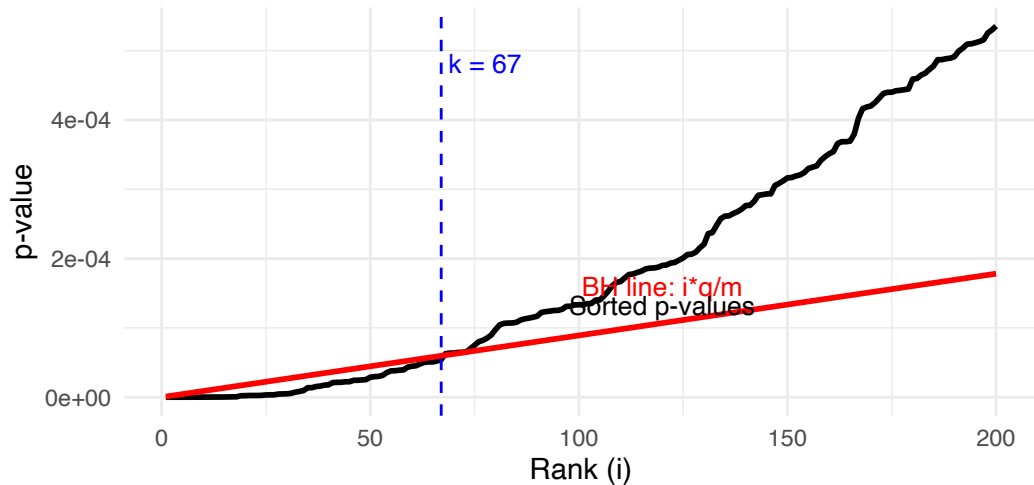
How did we get the “adjusted” p-value?

# Volcano plot: visualizing effect size and significance



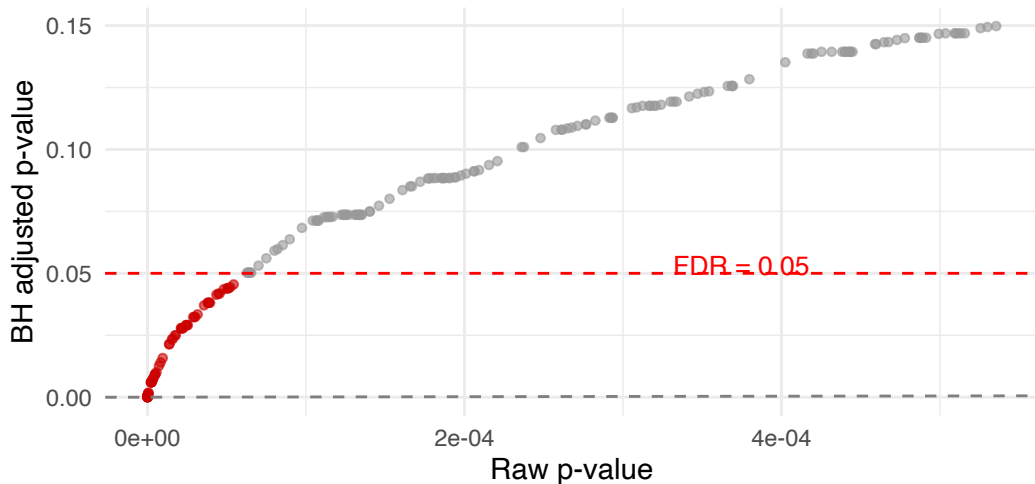
# Benjamini-Hochberg procedure visualized

Reject 67 hypotheses at  $FDR < 0.05$



# Benjamini-Hochberg procedure visualized

67 genes significant at  $FDR < 0.05$



## Recap: what are we controlling?

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**FWER** (Bonferroni, Holm): Controls **probability** of making **any** false discovery

$$\text{FWER} = P(V \geq 1) \leq \alpha$$

**FDR** (Benjamini-Hochberg): Controls **expectation** of **proportion** of false discoveries

$$\text{FDR} = E \left[ \frac{V}{R} \right] \leq q$$

### Key difference

BH controls the **expected** FDR, not the probability. In any single experiment, the actual FDR can be higher or lower than  $q$ .

# Today's lecture

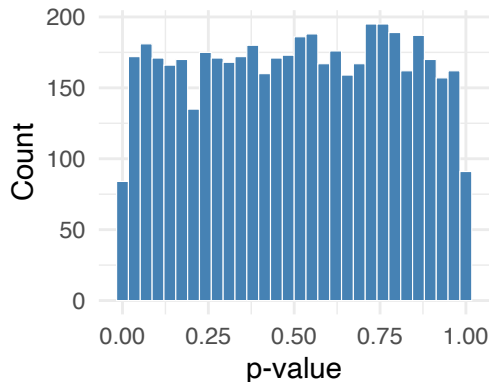
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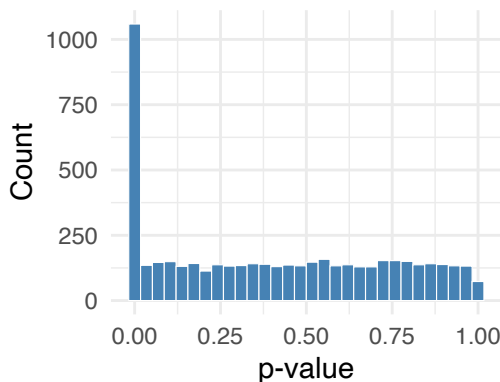
# Well-behaved p-values: uniform under the null

All multiple testing methods assume that p-values under  $H_0$  are **uniformly distributed**.

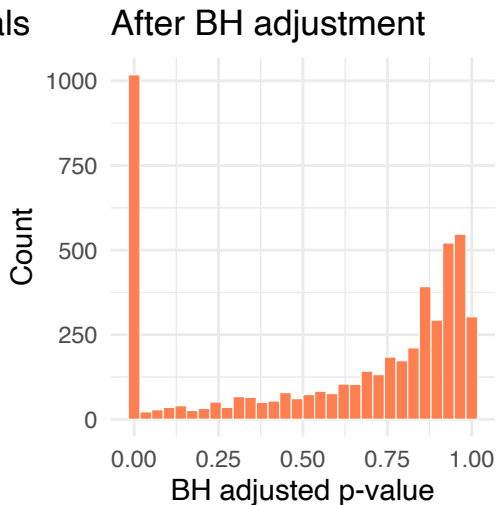
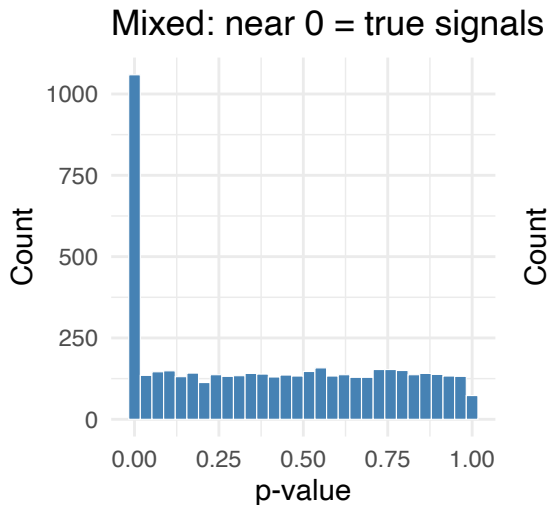
All null: uniform distribution



Mixed: near 0 = true signals



# BH adjusted p-value distribution



# Poorly behaved p-values

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When p-values **misbehave**, FDR estimates become invalid.

Common causes:

- Model misspecification (wrong distributional assumptions)
- Uncontrolled variation or batch effects
- Selection bias / data filtering after testing

Great reference:

<http://varianceexplained.org/statistics/interpreting-pvalue-histogram/>

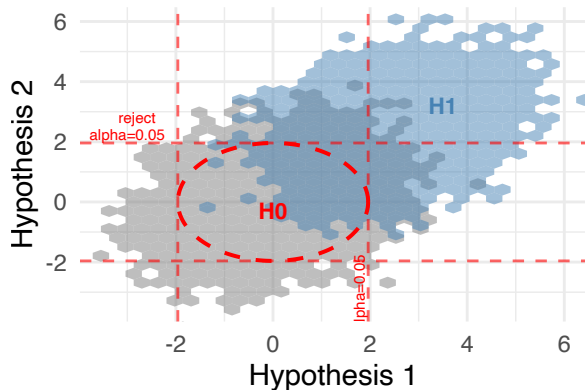
# What can we do to make sure that p-values

---

Possible solutions:

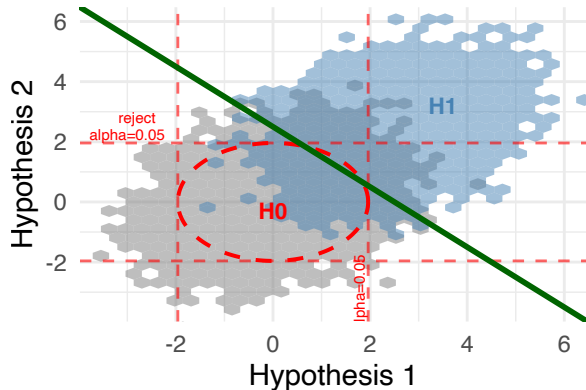
- **Nonparametric tests** (but limited flexibility for covariates)
- **Permutation testing**: shuffle group labels to construct an empirical null distribution; compare observed statistics to the permutation distribution
  - Limitation: computationally intensive; not ideal for very small sample sizes

# What can we estimate from data?



- Can we estimate the distribution under  $H_0$ ?
- Can we estimate the distribution under  $H_1$ ?
- Do we need to estimate what's going on under  $H_1$ ?

# What can we estimate from data?



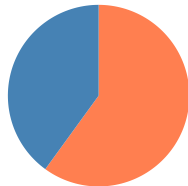
- Can we estimate the distribution under  $H_0$ ?
- Can we estimate the distribution under  $H_1$ ?
- Do we need to estimate what's going on under  $H_1$ ?

Can we come up with an *optimal* boundary?

# Bayesian view of multiple testing

**Mixture model framework:** Each hypothesis is either

- **Null** ( $H_0$ ) with prior probability  $\pi_0$
- **Alternative** ( $H_1$ ) with probability  $1 - \pi_0$



Status  Alternative  Null

**Bayesian interpretation** (note what's in the condition) (Storey 2003):

$$q(p_i) \approx P(H_0 \mid p \leq p_i)$$

# Bayesian view of multiple testing

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**Bayes rule:** 
$$P(H_0 \mid p \leq t) = \frac{P(p \leq t \mid H_0) \cdot P(H_0)}{P(p \leq t)}$$

$$P(H_0 \mid p \leq t) = \frac{t \cdot \pi_0}{\#\{p_i \leq t\}/m} \approx q(t)$$

(Storey 2003)

# Storey's q-value

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**Key insight:** BH assumes **all** hypotheses are null ( $\pi_0 = 1$ )

But in reality, many hypotheses are **truly alternative** (non-null)

**Storey's q-value:** Estimates  $\pi_0$  (proportion of true nulls) from data

Compare to BH:

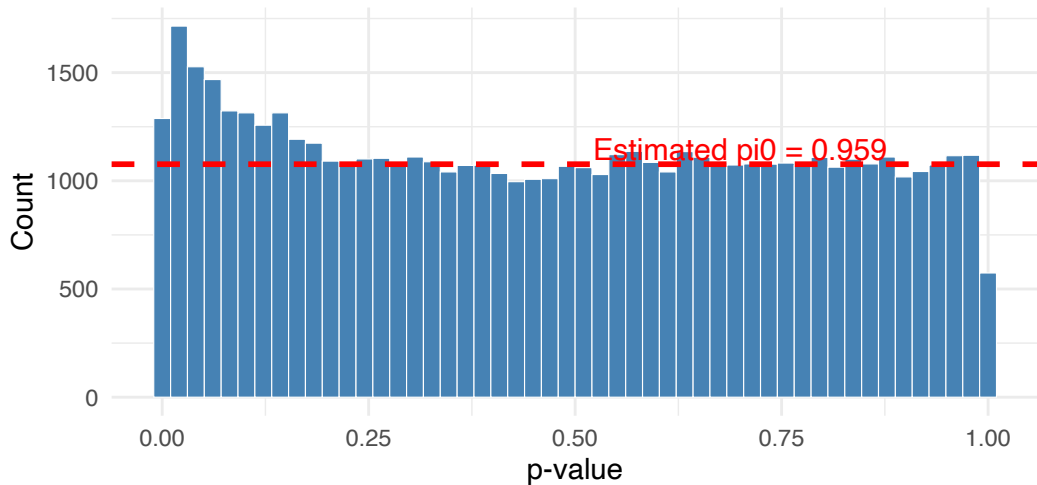
$$q(p_i) = \min_{j:p_j \geq p_i} \frac{\hat{\pi}_0 \cdot m \cdot p_j}{j}$$

$$\text{BH}(p_i) = \min_{j:p_j \geq p_i} \frac{m \cdot p_j}{j}$$

When  $\hat{\pi}_0 < 1$ : q-value is **less conservative** than BH

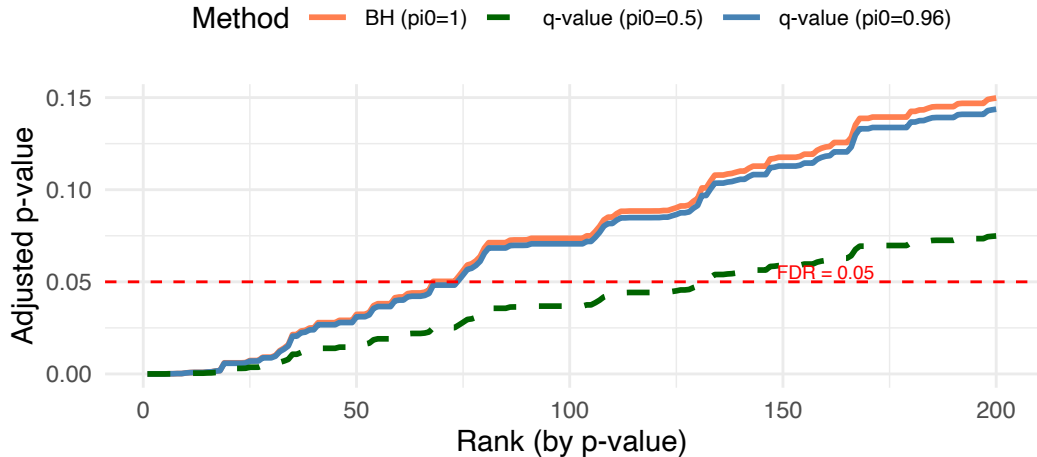
# Estimating $\pi_0$ from p-value histogram

P-value histogram: red line estimates null density

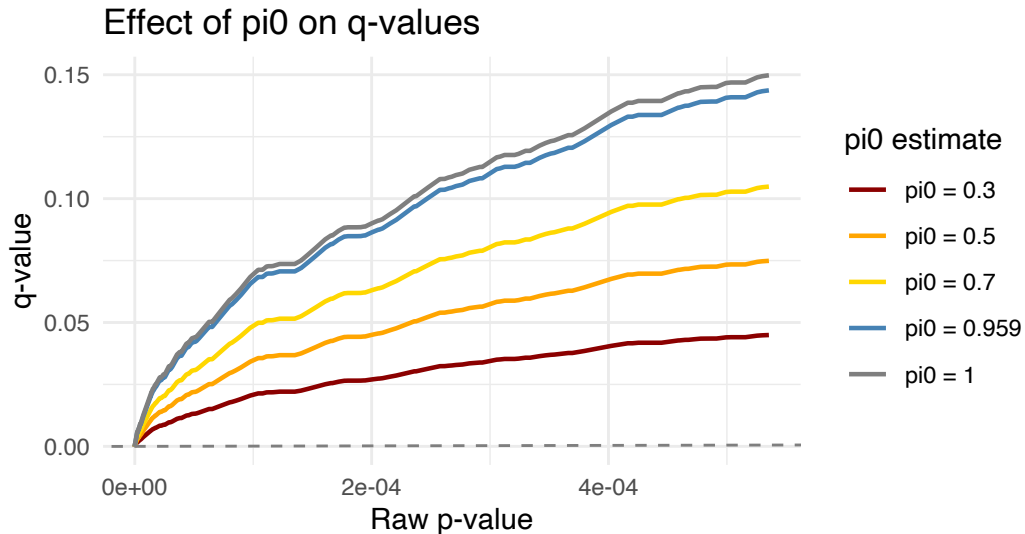


# Comparing BH vs. q-value

BH vs q-value with different  $\pi_0$



# Effect of different $\pi_0$ estimates



# When to use Storey's q-value vs. BH?

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## BH procedure:

- Conservative (assumes  $\pi_0 = 1$ )
- No assumptions about null proportion
- Valid (if independent)
- Recommended as default

## Storey's q-value:

- Less conservative when  $\pi_0 < 1$
- Estimates  $\pi_0$  from data
- More powerful when many true signals
- Risk:  $\pi_0$  estimation can be unstable with few tests

**Rule of thumb:** Use BH by default; consider q-value when you have  $> 1000$  tests with many expected signals

# Today's lecture

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- 1 Hypothesis testing in terms of two errors
- 2 Controlling Family-Wise Error Rate
- 3 False Discovery Rate
- 4 FWER, FDR, then what?
- 5 Beyond global FDR: gene-by-gene localized control

# Local false discovery rate (Efron et al. 2001; Efron 2010)

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**FDR:** Expected proportion of false discoveries **among all rejections**

**Local fdr:** Probability that **a specific test** is null

$$\text{lfdr}(z_i) = P(H_0 \mid z_i) = \frac{\pi_0 f_0(z_i)}{f(z_i)}$$

**Numerator:**  $\pi_0 f_0(z_i)$

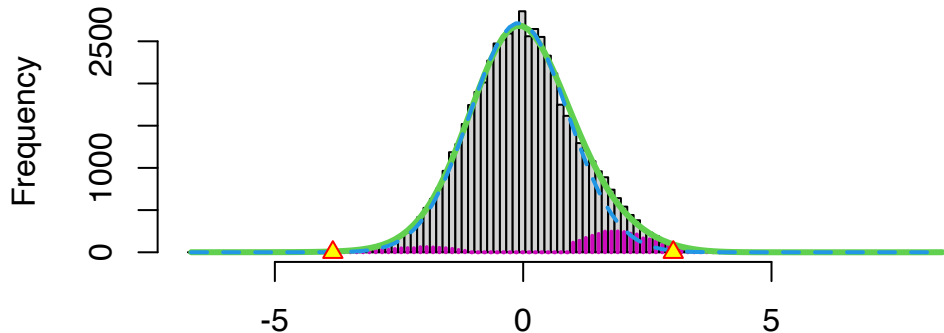
- Null contribution at  $z_i$

**Denominator:**  $f(z_i)$

- Total density at  $z_i$

Can estimate from empirical distribution of z-scores

# Local fdr: DEG example



MLE: delta: -0.085 sigma: 0.985 p0: 0.932

CME: delta: -0.074 sigma: 1.02 p0: 0.954

# q-value vs. local fdr

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## q-value (Storey)

$$q(p_i) = \min_{j: p_j \geq p_i} \frac{\pi_0 \cdot m \cdot p_j}{j}$$

- Based on **p-values**
- Ranks tests by p-value
- Minimum FDR at threshold  $p_i$
- **Tail probability** interpretation

## Local fdr (Efron)

$$\text{lfdr}(z_i) = \frac{\pi_0 f_0(z_i)}{f(z_i)}$$

- Based on **test statistics** (z-scores)
- Density ratio at specific  $z_i$
- Posterior  $P(\text{null} \mid z_i)$
- **Point-wise** interpretation

Both use  $\pi_0$ , but q-value uses **cumulative** p-values; local fdr uses **density** at each point

# What FDR doesn't tell us

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FDR controls the **expected proportion** of false discoveries

But for each gene, we want to know:

- Is this gene **truly differentially expressed**?
- What is the **direction** of the effect?
- What is the **magnitude** of the effect?

## The problem

A small p-value tells us there's an effect, but not its direction or size with certainty

# Local False Sign Rate (lfsr)

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**Local fsr:** Probability that the **sign** of an effect is wrong

$$\text{lfsr} = P(\text{sign}(\beta) \neq \text{sign}(\hat{\beta}) \mid \text{data})$$

**Key idea:** Account for uncertainty in **both**:

- Whether there's an effect ( $\beta \neq 0$ )
- The direction of the effect (sign of  $\beta$ )

# Adaptive Shrinkage (ashr)

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**ashr** (Stephens 2017): Empirical Bayes approach

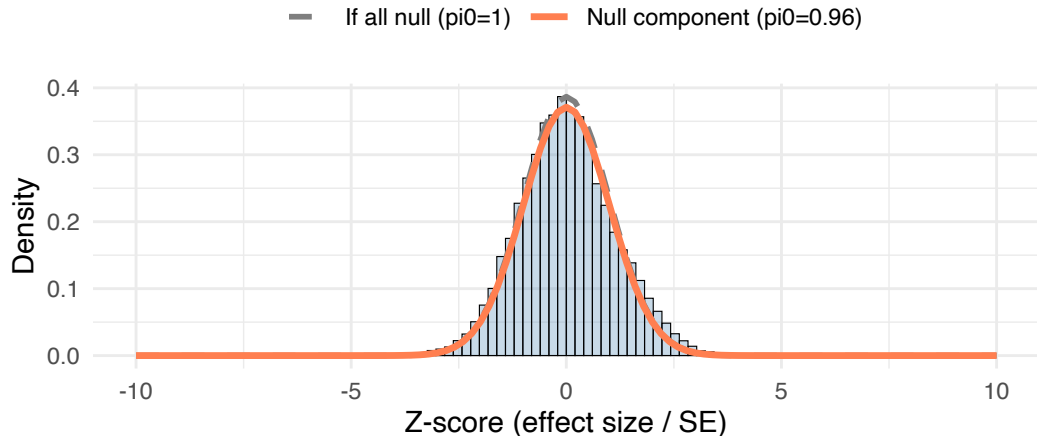
## Key features:

- ① Estimates effect size distribution from data
- ② Shrinks estimates toward zero (like in ridge regression)
- ③ Provides posterior probabilities for sign and magnitude
- ④ Computes local false sign rate (lfsr)

More conservative than FDR for **directional claims**

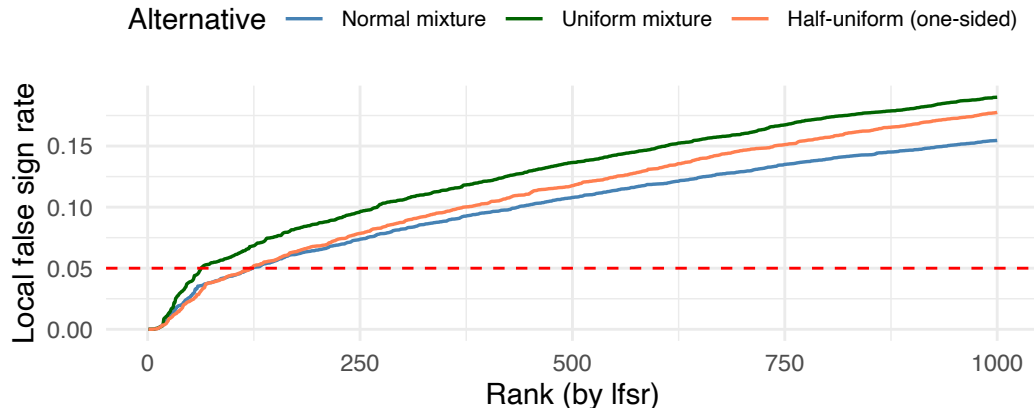
# ashr: Estimating null vs alternative distributions

## Observed vs null distribution



# ashr: Different alternative distributions

## Effect of alternative distribution assumption



Note: qvalue estimates  $\pi_0=0.96$  from p-values; ashR can give different estimates from effect sizes

# Why shrinkage?

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**Problem:** Raw effect size estimates are **noisy**

**Genes with large SE:** - Less precise measurements - Estimates unreliable - May be spuriously large

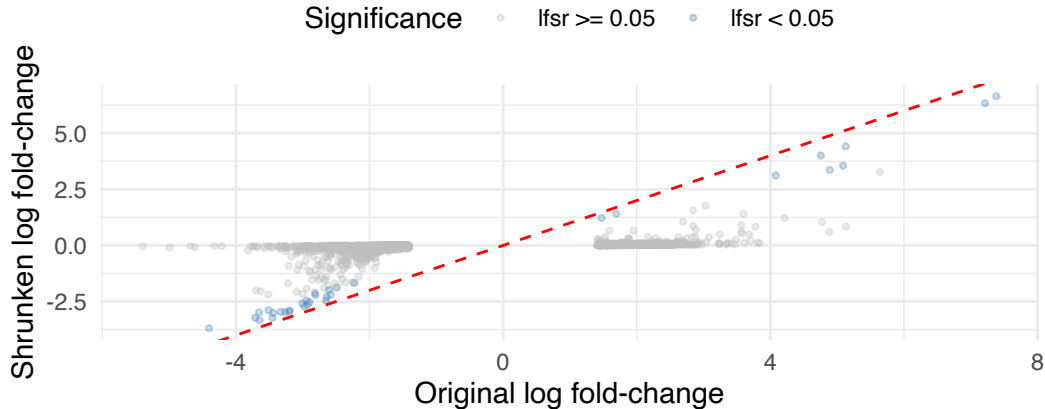
**Genes with small SE:** - More precise measurements - Estimates more reliable - Should be trusted more

**Solution:** **Shrink** uncertain estimates toward zero

- Small/noisy effects  $\rightarrow$  shrunk more
- Large/precise effects  $\rightarrow$  preserved
- **Borrows strength** across all genes

# ashr in action: Effect size shrinkage

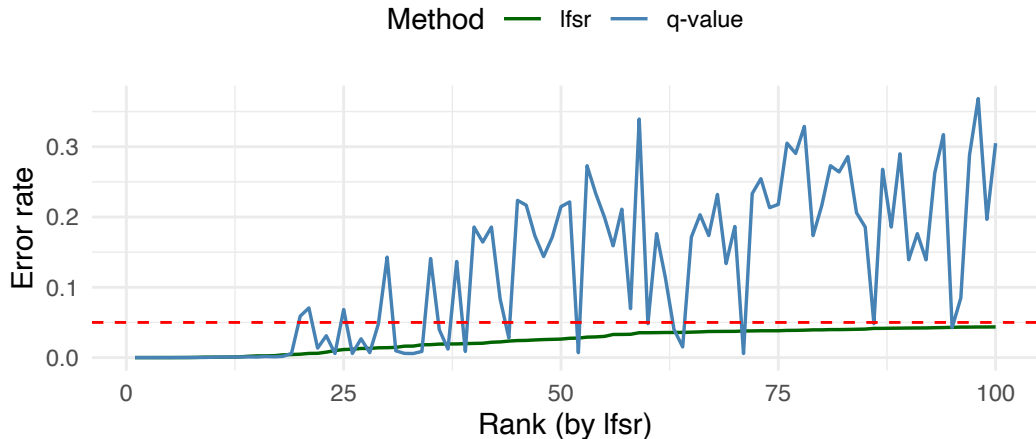
## Effect size shrinkage by ash



Points below diagonal: shrunk toward zero (less confident). Points near diagonal: preserved (more confident).

# ashr: local false sign rate vs q-value

## Local false sign rate vs q-value



# Summary: Multiple testing methods

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## Control FWER (strict):

- **Bonferroni:**  $\alpha/m$ 
  - Simple, conservative
  - Use for small  $m$  or independent tests
- **Holm:** Sequential Bonferroni
  - More powerful than Bonferroni
  - Still controls FWER

## Control FDR (flexible):

- **Benjamini-Hochberg:** Assumes  $\pi_0 = 1$ 
  - Standard FDR control
  - More discoveries than FWER
- **Storey q-value:** Estimates  $\pi_0$ 
  - More powerful when  $\pi_0 < 1$
  - Uses p-values only
- **ashr:** Uses effect sizes + SE
  - Accounts for measurement precision
  - `lfsr` for directions



# Reference

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- Benjamini, Yoav, and Yosef Hochberg. 1995. "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing." *J. R. Stat. Soc. Series B Stat. Methodol.* 57 (1): 289–300.
- Efron, Bradley. 2010. *Large-Scale Inference: Empirical Bayes Methods for Estimation, Testing, and Prediction*. Institute of Mathematical Statistics Monographs. Cambridge University Press.
- Efron, Bradley, Robert Tibshirani, John D Storey, and Virginia Tusher. 2001. "Empirical Bayes Analysis of a Microarray Experiment." *J. Am. Stat. Assoc.* 96 (456): 1151–60.

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Holm, Sture. 1979. “A Simple Sequentially Rejective Multiple Test Procedure.” *Scand. J. Stat.* 6 (2): 65–70.

Storey, John D. 2003. “The positive false discovery rate: a Bayesian interpretation and the q-value.” *Ann. Stat.* 31 (6): 2013–35.